Support Vector Machines

Objective In this assignment, you will implement core components of a Support Vector Machine (SVM) classifier. Your task is to compute the primal weight vector, identify support vectors, and detect slack vectors based on the SVM optimization problem. You will work with two provided datasets (kINSP and kSEP) and test your implementation on them.
Through this assignment, you will gain hands-on experience with key concepts in SVMs, including the decision boundary, margin, and slack. You will also practice using matrix operations and vectorized calculations.

Datasets We provide two datasets: `kINSP`: A set of labeled data points in a 2D feature space, and `kSEP`, another set of labeled data points in a 2D feature space. Each dataset contains three columns: the first two represent the feature values and the third column represents the class label.

Assignment Instructions

Task 1: Implement the Primal Weight Vector Calculation

In the function `weight_vector(x, y, alpha)`, you will compute the primal weight vector w. This vector defines the orientation of the decision boundary.

Task 2: Identify the Support Vectors

In the function `find_support(x, y, w, b)`, implement logic to find the support vectors–points that lie exactly on the margin; use a tolerance value to account for numerical imprecision.

Task 3: Find Slack Vectors

In the function `find_slack(x, y, w, b)`, detect points that violate the margin constraint in the soft-margin SVM formulation. A slack vector is either within the margin or misclassified.

Evaluation

- Correct computation of weight vector (33%)
- Correct identification of support vectors (33%)
- Correct identification of slack vectors: (33%)

You will be graded based on the correctness of your implementation. For this assignment, you do **not** need to provide any report required for "Theory to Practice Assignments."

Submission Guideline

Submit a zip file named `NAME_STUDENTID.zip` containing:

- `svm.py`: add a comment with your name at the top of this file.

Submit your zip file through the link on Blackboard. Submissions elsewhere won't be considered.